

CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
Higher 2

ANSWERS

BIOLOGY

9747/01

Paper 1 Multiple Choice

Friday 18 September 2009
1 hour 15min

Additional materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write and/or shade your name, NRIC / FIN number and HT group on the answer sheet in the spaces provided.

There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Calculators may be used

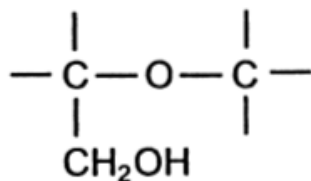
This document consists of **15** printed pages.

[Turn over]

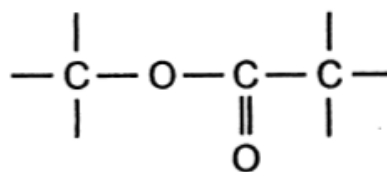
1.1] Biological Molecules1-2

1 Which set of bonds is found in triglycerides?

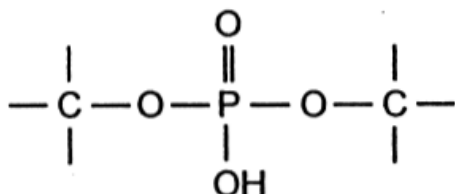
A



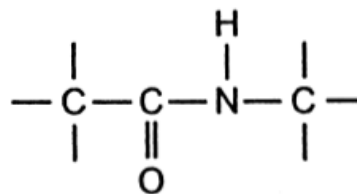
B



C

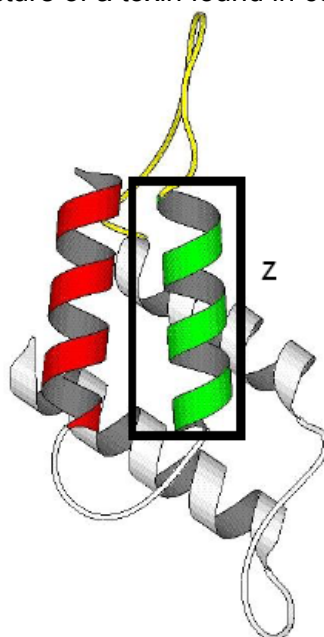


D



Answer B

2 The diagram below shows the structure of a toxin found in cobra venom.



Which correctly describes the structure of the part enclosed in box Z?

Number of Amino Acids Present

Types of Bonds Present

- A Cannot be determined
- B Cannot be determined
- C Approximately 11 amino acids
- D Approximately 11 amino acids

- Hydrogen Bonds only
- Hydrogen Bonds and hydrophilic bonds
- Hydrogen bonds only
- Hydrogen bonds and hydrophobic bonds

Answer C

Ans: 3.6a.a. per turn, only H bonds

1.2] Cell Functions 3

3 The table gives descriptions of four membranous structures in a cell.

Which structure is correctly matched with its function?

	structure	function
A	an extensive network of tubes and sacs; each tube and sac bounded by a single membrane	lipid synthesis
B	a spherical sac bounded by a single membrane	protein synthesis
C	a sac bounded by two membranes, the inner highly folded	packaging of proteins
D	a stack of elongated, curved sacs; each sac bounded by a single membrane	photosynthesis

Answer: A

Exm Report 07H2P1Q3 While this was not a difficult question, option D was attractive to some weaker candidates who were perhaps not familiar with the description of smooth endoplasmic reticulum (option A). A stack of sacs suggests a granum of a chloroplast, but they are described as curved, which identifies the structure as a Golgi body.

1.3] Enzymes 4-5

4 A 2 % solution of starch (amylose) and a 2 % solution of amylase were mixed together and incubated at 25°C. After one hour, samples of the solution were tested with three reagents.

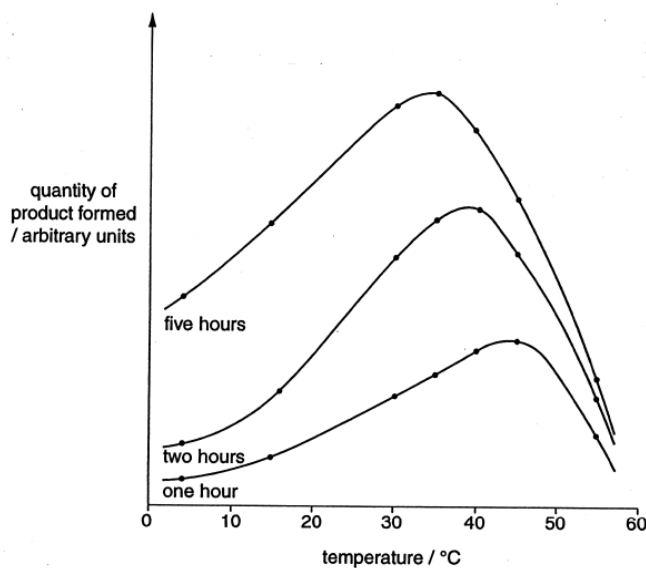
Which result would be expected?

	reagent added to samples		
	biuret solution	Benedict's solution and heated	iodine in potassium iodide solution
A	blue	brick red	brown
B	blue	blue	blue-black
C	purple	blue	blue-black
D	purple	brick red	brown

Answer: D

Exm Report 04H2P1Q8 Almost all candidates recognised that there would be a negative result for starch and a positive one for reducing sugar but a majority did not take into account that the enzyme is protein, which, at a concentration as high as 2%, would give a positive result with biuret solution.

- 5 The graph shows the results of an experiment in which samples containing the same concentration of enzyme and substrate were kept at different temperatures for periods of one, two and five hours. The quantities of product formed were then determined.



Which of the following options best explains why the optimum temperature is lower if the quantity of product formed is measured after five hours rather than one hour?

- A Tertiary bonds are not broken at higher temperatures.
- B The enzyme has a range of optimum temperatures.
- C A longer time at high temperature denatures enzyme.
- D The optimum temperature for the enzymes is 45 °C.

Answer: C

2006 A level Qn 2 modified -enzymes would be a longer time at high temperatures and that this would lead to increasing denaturation with more enzyme denatured due breaking hydrogen bonds and loss of tertiary structure and shape of active site.

2.2] Telomerase 6

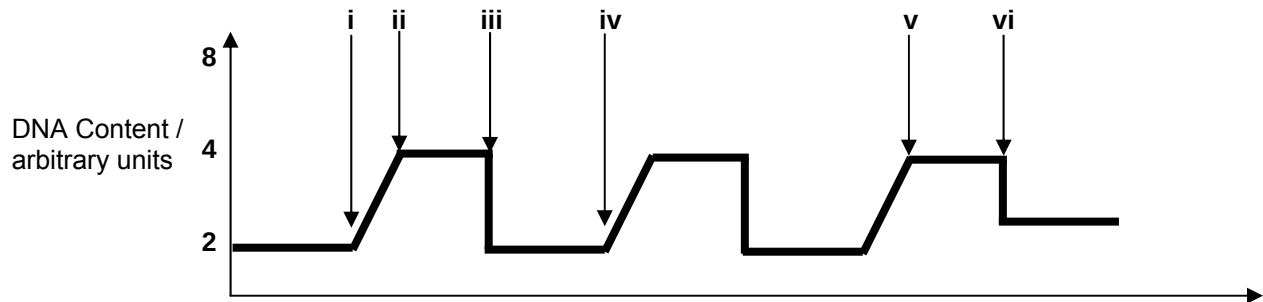
- 6 What describes common features of the telomere and the centromere?
- A Both areas have proteins associated with them.
 - B Both are made up of DNA with a high incidence of adenine and thymine.
 - C Both DNA sequences are conserved throughout the life of the organism.
 - D During interphase DNA replication takes place simultaneously in both regions.

Answer A

Exm Report 07H2P1Q15 Many candidates chose option B and low scoring candidates were also attracted to option C. This suggests that not all candidates understand telomeres and centromeres.

2.3] Mitosis 7

7 The figure below shows DNA content of a cell undergoing *abnormal* cell division.



Which one of the following best describes what is happening at stages i to v?

	i to ii	iii	iii to iv	v to vi
A	non disjunction	Meiosis I	G2 phase	meiosis II
B	fusion of gametes	mitosis	S phase	mitosis
C	S phase	cytokinesis	G1 phase	non disjunction
D	non disjunction	cytokinesis	G2 phase	mitosis

Answer: C

2.4] Meiosis 8

8 Which of the following statements about meiosis is **false**.

- A Sister chromatids are separated at anaphase II.
- B Homologous chromosomes are separated in anaphase I.
- C Cells at the beginning of Meiosis II are haploid.
- D Homologous chromosomes are paired on the metaphase plate in metaphase I.

Answer: A

Due to crossing over in Prophase I there are non-sister chromatids separating in anaphase II.

3.1] Replication 9-10

9 Pyrimidine bases contain 4 carbon atoms and purine bases contain 5.

How many carbon atoms are there in a nucleotide containing uracil?

- A 4
- B 5
- C 9
- D 10

Answer: C

Exm Report 06H2P1Q16 Many candidates were attracted to option A, suggesting a misreading of the question such that only the number of carbon atoms in uracil was counted.

Uracil = Pyrimidine (4C) + Ribose (5C)=9C

10 Which of the following statements about replication is **false**?

- A No ATP is utilized in the formation of phosphodiester bonds on the leading strand.
- B DNA polymerase I and III both needs a free 3'OH to carry out their function.
- C Synthesis of the lagging strand occurs in a 3' to 5' direction.
- D On the lagging strand, Primase is hindered by action of Helicase thus forming Okazaki fragments.

Answer: C

3.2] Transcription 11-12

11 The base analogue 5-bromouracil is a chemical mutagen that induces single-pair substitution in DNA called transitions (i.e. substitutions of one purine for another purine or one pyrimidine for another pyrimidine).

The genetic code (the code is given here in terms of the mRNA codons)

		second base							
		U		C		A		G	
first base	U	UUU } Phe	UUC }	UCU } Ser	UCC }	UAU } Tyr	UAC }	UGU } Cys	U
		UUA } Leu	UUG }	UCA }	UCG }	UAA } stop	UAG }	UGA } stop	A
						UAG } stop		UGG } Trp	G
C		CUU } Leu	CUC }	CCU } Pro	CCC }	CAU } His	CAC }	CGU } Arg	U
		CUA } Leu	CUG }	CCA }	CCG }	CAA } Gln	CAG }	CGA }	C
								CGG }	A
									G
A		AUU } Ile	AUC }	ACU } Thr	ACC }	AAU } Asn	AAC }	AGU } Ser	U
		AUA } Met	AUG }	ACA }	ACG }	AAA } Lys	AAG }	AGA }	C
								AGG }	A
									G
G		GUU } Val	GUC }	GCU } Ala	GCC }	GAU } Asp	GAC }	GGU } Gly	U
		GUA } Val	GUG }	GCA }	GCG }	GAA } Glu	GAG }	GGG }	C
									A
									G

Using the genetic code provided, which one of the following amino acid substitutions would you expect to be induced by 5-bromouracil with the highest frequency?

- A Met to Val
- B Met to Leu
- C Lys to Thr
- D Lys to Gln

Answer: A

12 The table shows the percentages of nitrogenous bases in four samples of nucleic acids.

Which base is adenine?

Sample	Bases				
	A	B	C	D	Uracil
1	19	31	30	19	nil
2	27	23	24	26	nil
3	25	25	nil	25	25
4	17	32	33	18	nil

Answer: B

From 1 DNA A=D B=C
 From 2 DNA A=D B=C
 From 3 RNA C=T
 From 4 DNA A=D B=C

Therefore B is Adenine

3.3] Translation 13

13 What is the minimum number of GTP and ATP used in the synthesis of a pentapeptide?

	GTP	ATP
A	6	0
B	11	5
C	0	16
D	6	5

Answer: B

A.A	A1 +A2	A1A2+A3	A1A2A3+A4	A1A2A3A4+A5		
Stage:	Initiation	Translocation1	Translocation2	Translocation3	Translocation4	Termination
GTP	x1	x2	x2	x2	x2	x2
ATP	1xATP for each activated A.A.					
						Total: 11 GTP
						Total: 5 ATP

4.1] Cell signaling 14-15

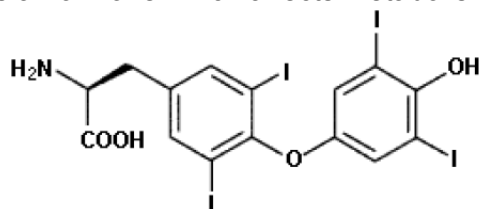
14 Which of the following statements are true about the signal transduction pathways?

- 1 They enable different cells to respond appropriately to the same signal.
- 2 They help cells use up phosphate generated by ATP breakdown.
- 3 They can amplify a signal.
- 4 Variations in the signal transduction pathways can enhance response specificity.

- A 1, 2 and 3
 B 1, 3 and 4
 C 2 and 4 only
 D 1 and 3 only

Ans : B

15 The diagram shows a hormone which affects metabolism in humans.



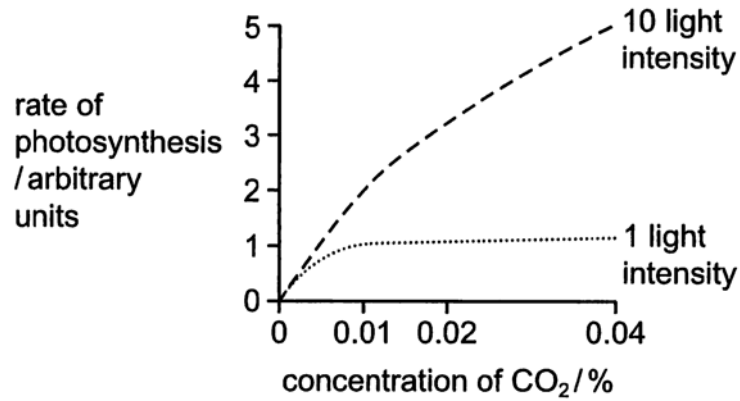
How does this hormone act on its target cell?

- A It activates an enzyme cascade that amplifies the hormone signal.
 B It alters specific receptor sites on the cell surface membrane.
 C It enters the cell and binds to nuclear receptors.
 D It inhibits the synthesis of cholesterol molecules.

Ans : C

2T26

16 The graph shows the effect of carbon dioxide concentration on photosynthesis at two different light intensities.

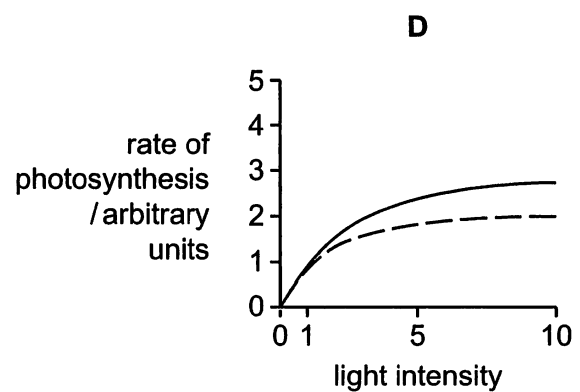
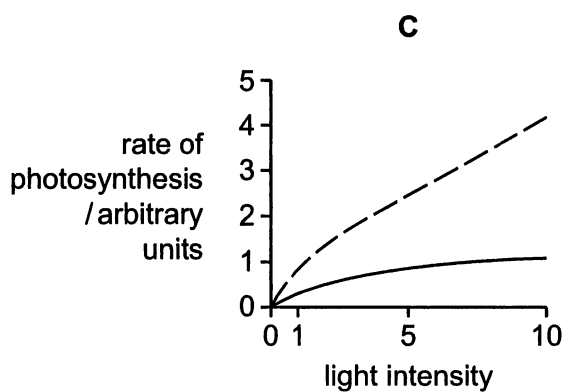
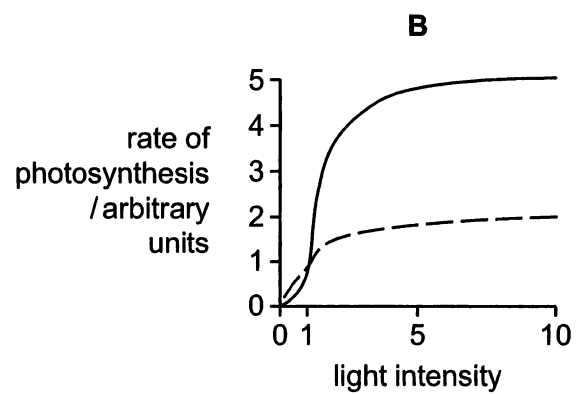
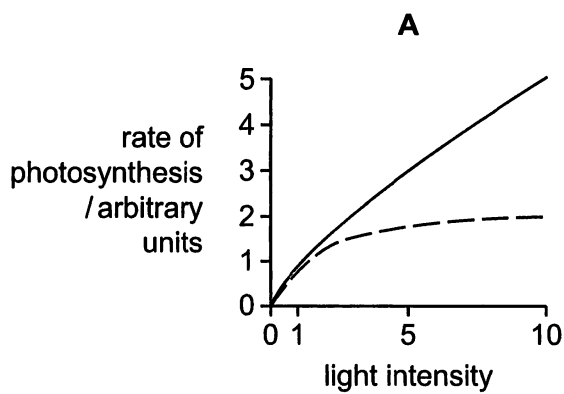


Which graph correctly shows the effect of light intensity on the rate of photosynthesis, at two different carbon dioxide concentrations?

key

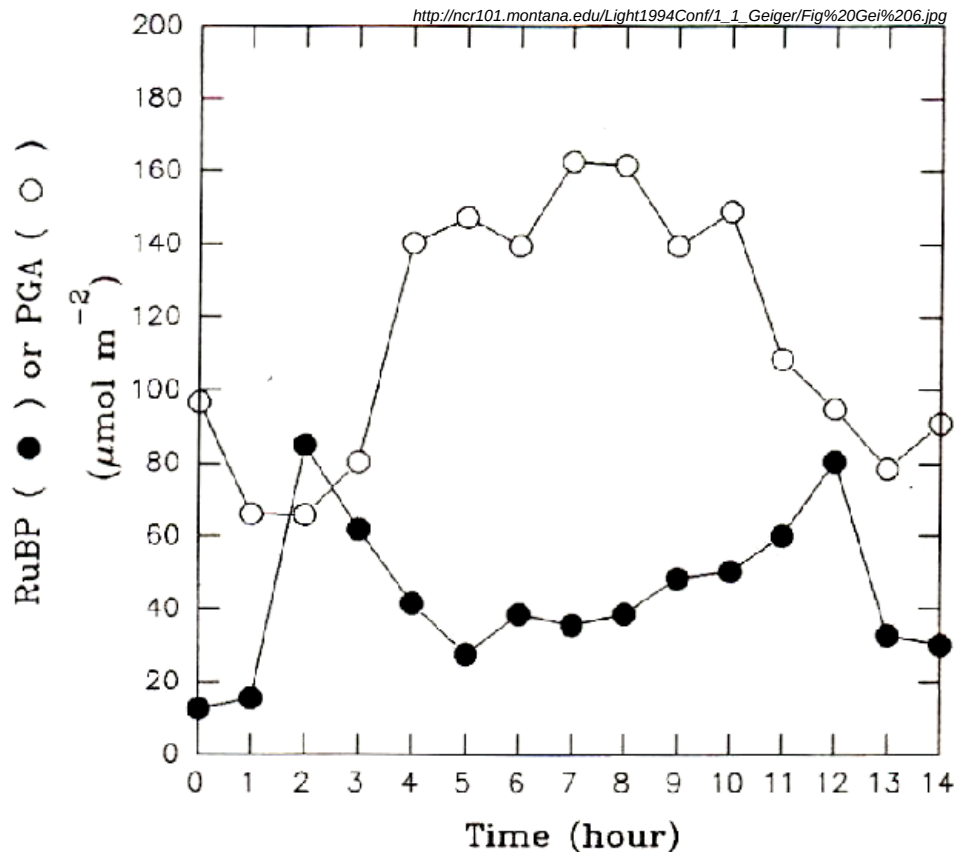
--- 0.01 % concentration of CO₂

— 0.04 % concentration of CO₂



Answer: A
07H2P1Q26

17 The figure below shows the time courses of RuBP and PGA levels in a sugar beet leaf during a 14-h light regime that simulates a natural day. (Ref: From *Servaites et al.*, 1989a)



With reference to the figure above, which one of the following statements is not true.

- A Time 1-2hr: The rise of RuBP coincides with the light no longer being the limiting factor.
- B PGA levels are highest at Time 7hrs when photosynthesis is at its highest rate.
- C RuBP rises steadily from Time 5-12hr due to regeneration from PGA directly.
- D CO_2 levels are on the rise as stomata open between Time 2-7hr.

Answer: C

regeneration of RuBP is from Triose Phosphate directly not PGA.

2T25

4.4] Respiration 18-19

18 During aerobic respiration ATP can be formed by oxidative phosphorylation in the electron transport system and or by substrate-level phosphorylation. In the complete oxidation of one molecule of glucose, approximately what percentage of ATP is formed by substrate-level phosphorylation?

- A 10%
- B 11%
- C 8%
- D 15%

Answer D

NOTE: Total ATP formed
Substrate level Phosphorylation

=40ATP (including 2ATP invested at start of Glycolysis)
Glycolysis **Kreb (Between Succinal CoA and Succinate)**
 2ATP x2(GALP) 1xATP x2(Succinate formation)
 =6 ATP
 = 6/40 = 15%

19 Which changes occur when the electron transfer system in animals is inhibited?

	lactate concentration	oxygen consumption	Krebs cycle activity
A	decreased	decreased	increased
B	decreased	increased	increased
C	increased	decreased	decreased
D	increased	increased	decreased

Answer C

4.5] Hormonal Control 20

20 Which of the following homeostatic control mechanisms is correct?

	Hormone / Ligand	Receptor mechanism	Physiological effect
A	Insulin	G-Protein linked receptor	decreased glycogenesis
B	Insulin	Tyrosine Kinase Receptor	increased gluconeogenesis
C	Glucaon	G-Protein linked receptor	increased gluconeogenesis
D	Glucaon	Tyrosine Kinase Receptor	decreased glycogenesis

Answer: C

glycogenesis- conversion Glu to Glycogen

gluconeogenesis- conversion of aa and Lipids to Glu

4.7] Nervous control 21-22

21 Curare causes paralysis of striated muscle and is used during general anesthesia.

When curare is applied to a nerve-muscle preparation, the muscle does not contract when the nerve is stimulated, but if a stimulus is applied directly to the muscle, contraction occurs.

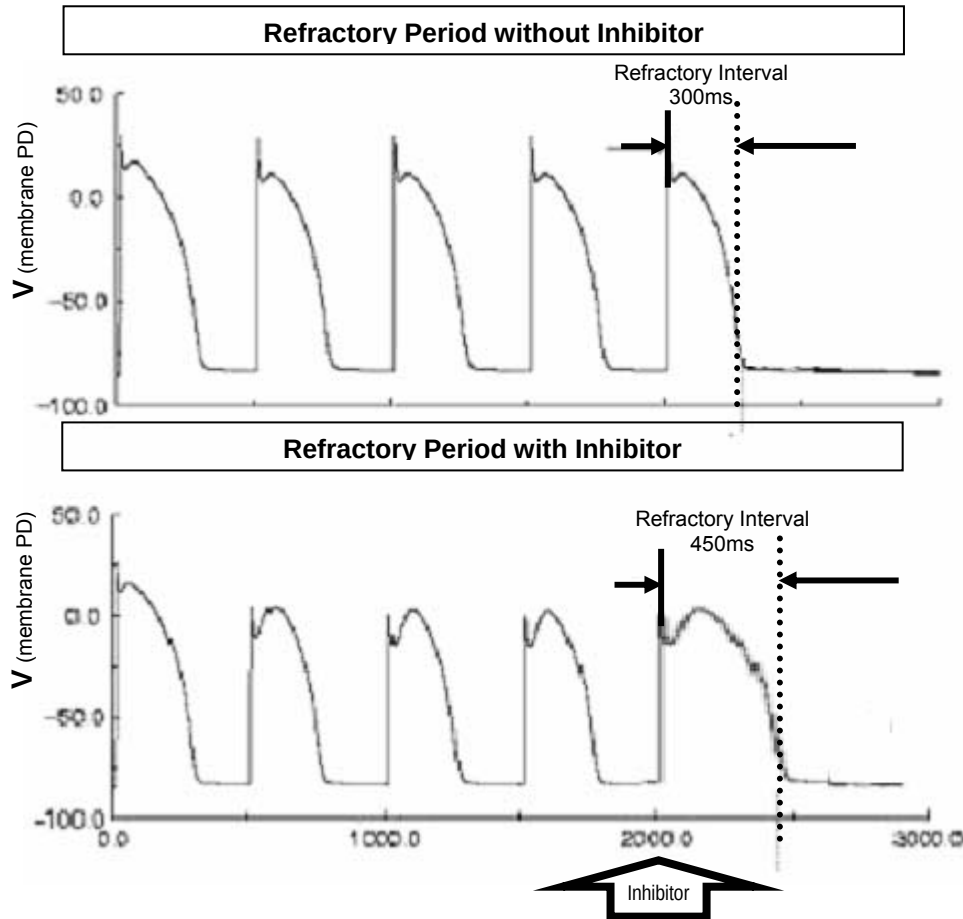
When curare is applied to the nerve only, the muscle contracts when the nerve is stimulated.

Which of the following statements correctly explains the effect of curare?

- A** It affects the formation of the enzyme cholinesterase.
- B** It inhibits the development of an action potential in the nerve fibre.
- C** It prevents the depolarization of the post-synaptic membrane.
- D** It stimulates the formation of acetylcholine at the motor end plate.

Answer: C NJC 07PrelimQn 33

22 The figure shows the effects of a nerve inhibitor on refractory intervals in mice neurons.



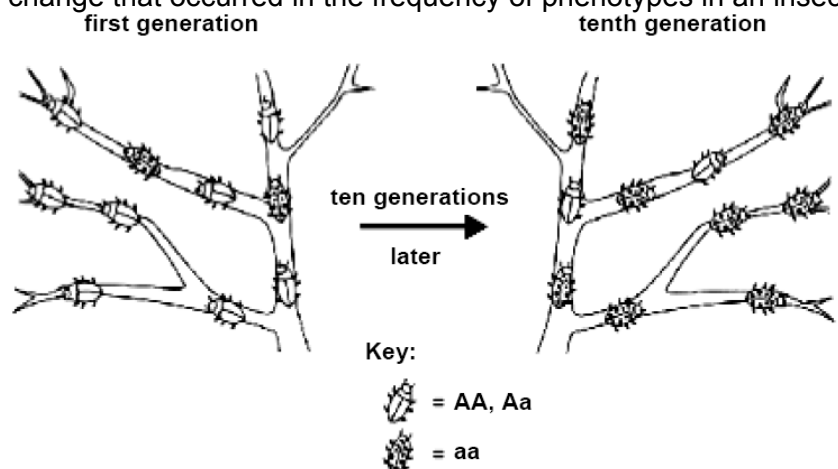
With reference to the figure above, what can be concluded about the inhibitor.

- A It inhibits the action of the Na^+/K^+ pump therefore prolonging the refractory interval.
- B It inhibits the development of an action potential by prolonging the closure of Na^+ gated channels.
- C It inhibits ATPase in its action.
- D It inhibits the exit of K^+ ions out of the Neuron

Answer: B

5.2] Natural Selection & Evolution & Variation 23-24

23 The diagram below illustrates the change that occurred in the frequency of phenotypes in an insect population over ten generations.



Which of the following is a probable explanation for this change?

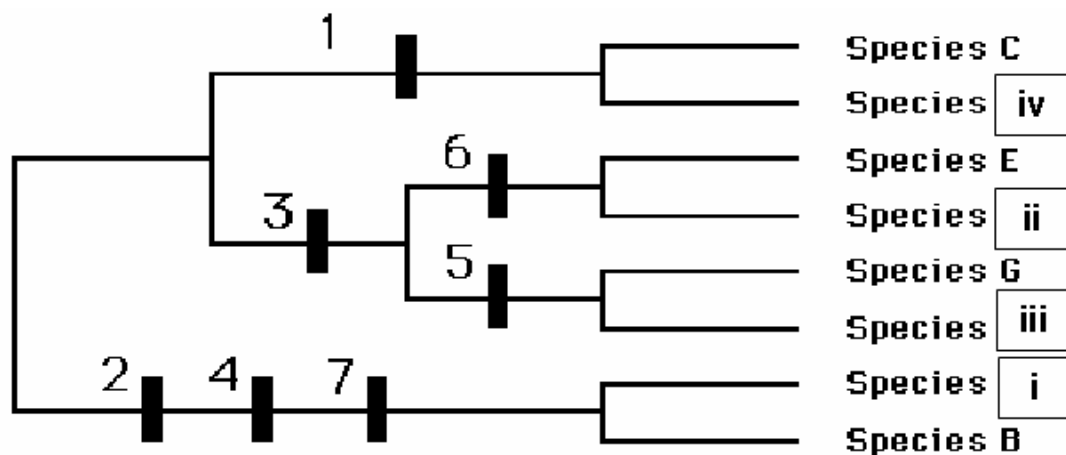
- A Over time, there was a decrease in the adaptive value of allele a.
- B Over time, there was an increase in the adaptive value of allele a.

- C Over time, there was an increase in the population of this insect.
- D Over time, there was a decrease in the mutation rate of allele A.

Answer: B NJC 07Prelim Qn40

24 The figure below shows the phylogeny of eight species based on the similarities in DNA sequences of a specific gene. The DNA sequence is divided into seven sections to enable easier comparison. Black bars on the horizontal lines indicate the similarities in the different species on each branch.

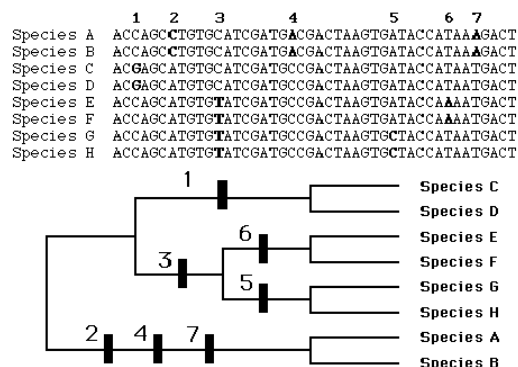
Spp			1				2		3				4							5			6		7			
A	A	C	C	A	G	C	C	T	C	A	T	C	G	A	T	G	A	C	T	A	T	G	A	T	A	A	G	
B	A	C	C	A	G	C	C	T	C	A	T	C	G	A	T	G	A	C	T	A	T	G	A	T	A	T	A	G
C	A	C	G	A	G	C	A	T	C	A	T	C	G	C	T	G	A	C	T	A	T	G	A	T	A	T	A	G
D	A	C	G	A	G	C	A	T	C	A	T	C	G	C	T	G	A	C	T	A	T	G	A	T	A	T	A	G
E	A	C	C	A	G	C	A	T	T	A	T	C	G	C	T	G	A	C	T	A	T	G	A	T	A	A	A	G
F	A	C	C	A	G	C	A	T	T	A	T	C	G	C	T	G	A	C	T	A	T	G	A	T	A	A	A	G
G	A	C	C	A	G	C	A	T	T	A	T	C	G	C	T	G	A	C	T	A	T	G	C	T	A	T	A	G
H	A	C	C	A	G	C	A	T	T	A	T	C	G	C	T	G	A	C	T	A	T	G	C	T	A	T	A	G



Which of the following is correct?

	i	ii	iii	iv
A Species	D	H	F	A
B Species	D	F	H	A
C Species	A	F	H	D
D Species	A	H	F	D

Ans: C



5.3] Evolution 25

25 Which of the following ideas were **not** part of Darwin's theory of evolution by natural selection?

I	Acquired variation is inherited.
II	Variation between individuals arises by gene mutation.
III	Organisms produce more offspring than the environment can support.
IV	Only those individuals best adapted to the environment survive and reproduce.
V	Regions that encode portions of the polypeptide that are vital for structure and function are less likely to incur mutations.

- A I, II and V
 B I, III and IV
 C II, III and V
 D II, IV and V

Answer: A

HCI Q34 Prelim 2008

6.2] Mutation 26

26 The table shows the mRNA codons for 11 different amino acids.

amino acid	mRNA codon
Ala	GCG
Glu	GAG
His	CAC
Leu	CUG CUC

amino acid	mRNA codon
Lys	AAG
Pro	CCU
Thr	ACU
Val	GUG

amino acid	mRNA codon
Arg	CGC
Phe	UUC
Gly	GGA

The first seven DNA triplets coding for a protein are shown below. These are DNA triplets from the coding strand, complementary to the mRNA.

CACGAGGCGAAGGGACACCGC

A mutation occurs when the seventeenth nucleotide in the DNA sequence is changed to G.

What is the amino acid sequence for the mutation section of DNA?

- A His Glu Ala Lys Gly Arg Arg
 B Val Leu Arg Phe Pro Ala Ala
 C His Glu Ala Lys Gly His Arg
 D Val Leu Arg Phe Pro Val Ala

Ans B Nov 07 Q9

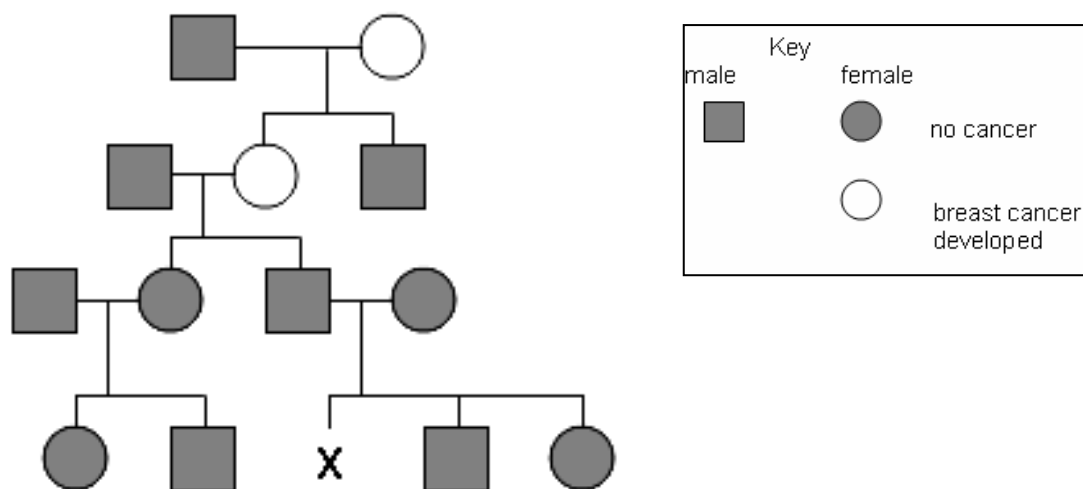
6.3] Di n Mono Hybrid Crosses 27-28

27 In *Drosophila*, ebony body colour and vestigial wings are recessive to the wild type and are not linked. Crossing two flies, heterozygous for both characters, produced 256 offspring. How many offspring would be expected to have ebony body colour and normal wings?

- A 16
- B 48
- C 96
- D 144

Ans: B Nov 00 Q23

28 The diagram shows the inheritance of a form of breast cancer associated with the presence of just one allele of the autosomal gene BRCA 1.



What is the probability that woman **X** inherits the BRCA 1 allele associated with breast cancer?

- A 0.00
- B 0.25
- C 0.50
- D 1.00

Ans: B Nov 01 Q20

Male can be a carrier without developing breast cancer.

$P(\text{male has BRCA 1}) = 0.5$

$P(\text{allele passed on to daughter}) = 0.5$

Therefore, $P(\text{woman X inherits BRCA 1}) = 0.5 \times 0.5 = 0.25$

6.5] Interactions between genes 29

29 Agouti mice have banded hairs, giving a grey colour. Black mice have unbanded hairs. White mice have no pigment. A cross between a homozygous black mouse and a white mouse produced offspring with agouti hair. Another cross between the black mouse and another white mouse produced some offspring with agouti hair and some with black hair.

What explains these observations on the phenotype of hair of mice?

- A There is a single gene with two codominant alleles, black and white.
- B There is a single gene with three alleles in a dominance series, black → grey → white.
- C There are two epistatic genes, one controlling pigment production and one controlling banding.
- D There are two linked genes, one controlling pigment production and one controlling banding.

Ans: C Nov 07 Q20

- 30** A tall, green-stemmed plant with the genotype **TTrr** was crossed to a short, red-stemmed plant with the genotype **ttRR**. The F_1 plants were allowed to self fertilise. A χ^2 test was carried out on the results obtained for the F_2 generation.

Part of the table of values for χ^2 is shown.

degrees of freedom	p = 0.5	p = 0.1	p = 0.05	p = 0.01	p = 0.001
1	0.46	2.71	3.84	6.64	10.83
2	1.39	4.6	5.99	9.21	13.82
3	2.37	6.25	7.82	11.34	16.27
4	3.36	7.78	9.49	13.28	18.46
5	4.35	9.24	11.07	15.09	20.52

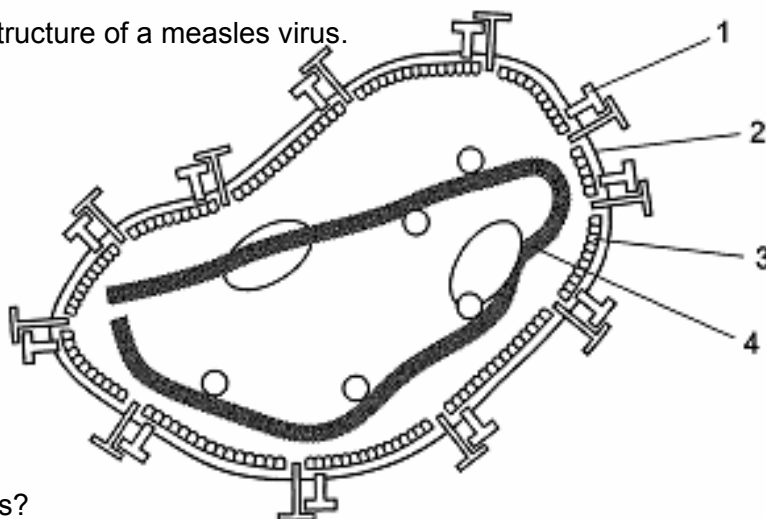
The value of χ^2 in this investigation was 7.6.

What is the probability of this value of χ^2 and do the results fit the expected ratio?

	probability	results fit expected ratio
A	between 0.001 and 0.01	no
B	between 0.01 and 0.05	yes
C	between 0.05 and 0.1	yes
D	between 0.1 and 0.5	no

Ans: C Nov 05 Q22

- 31** The diagram shows the structure of a measles virus.



What are the labeled parts?

	capsomeres	glycoprotein	lipid bilayer	RNA
A	1	3	4	2
B	1	3	2	4
C	3	1	2	4
D	3	1	4	2

Ans: C Nov 07 Q10

32 A virus has a base ratio of A + T : G + C = 4 : 3 : 2 : 1

What type of virus is this?

- A A single-stranded DNA virus.
- B A single-stranded RNA virus.
- C A double-stranded DNA virus.
- D A double-stranded RNA virus.

Ans: A (modified from AJC Prelim 08/P1/17)

7.2] Genetics of Viruses

33-34

33 Which correct describes the transfer of DNA from one bacterium to another?

	binary fission	transduction	conjugation
A	bacterial chromosome and plasmids pass to daughter cells	DNA transferred by bacteriophage from one bacterium to another	single strand of F plasmid transferred from one bacterium to another
B	bacterial chromosome and plasmids pass to daughter cells	bacterial cell takes up foreign DNA from culture medium	double strand of F plasmid transferred from one bacterium to another
C	only plasmids pass to daughter cells	DNA transferred by bacteriophage from one bacterium to another	single strand of F plasmid transferred from one bacterium to another
D	only plasmids pass to daughter cells	bacterial cell takes up foreign DNA from culture medium	double strand of F plasmid transferred from one bacterium to another

Ans: A Nov 08 Q12

34 In the absence of glucose and lactose from the medium, the *lac* operon in E.coli was actively transcribed.

What could have caused the above observation?

- 1 Operator of the *lac* operon is always bound to the repressor protein.
- 2 Presence of a mutation in the *lacA* gene.
- 3 There is a mutation in the operator.
- 4 Mutation in the promoter.

- A 1 and 3
- B 2 and 3
- C 3 only
- D 3 and 4

Ans: C CJC 09 Prelims

8.1] Structure of Eukaryotic chromatin 35

35 At prophase of mitosis, a eukaryote chromosome consists of two chromatids.

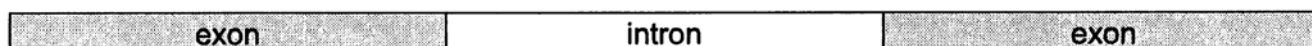
What is the structure of a single chromatid?

- A** one molecule of single-stranded DNA coiled around protein molecules
- B** two molecules of single-stranded DNA each coiled around protein molecules
- C** one double helix of DNA coiled around protein molecules
- D** two double helices of DNA each coiled around protein molecules

Ans: C Nov 99 Q11

8.3] Control of Gene Expression 36-37

36 The diagram shows a gene in a human cell.



In one form of the blood condition thalassaemia, there is a mutation in an intron of the haemoglobin gene.

Which stage of gene expression would this mutation directly affect?

- A** activation of the gene by a steroid hormone
- B** attachment of mRNA to a ribosome
- C** binding of RNA polymerase to DNA
- D** splicing of pre-mRNA

Which type of control of gene expression is commonly found in eukaryotes?

- A** post transcriptional
- B** post translational
- C** transcriptional
- D** translational

Ans: D Nov 07 Q16

37 Which type of gene control is commonly found in both prokaryotes and eukaryotes?

- A** Post transcriptional
- B** Post translational
- C** Transcriptional
- D** Translational

Ans: C Nov 08 Q18 Option A and option B were both attractive to low scoring candidates, suggesting there is some uncertainty about the sequence of processing of pre-mRNA

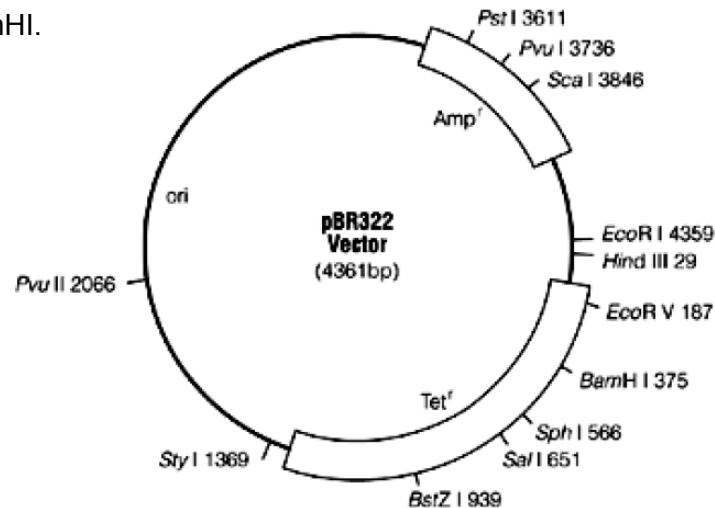
8.4] Molecular Biology of cancer 38

38 What is the normal function of tumour suppressor genes?

- A** to change the coding sequences in proto-oncogenes so that they become oncogenes
- B** to encode proteins that stimulate cell proliferation
- C** to prevent the stimulating activity of cellular oncogenes
- D** to repair any damage to the cell's DNA

Ans: C Nov 08 Q19

39 pBR322 vector is used to clone a eukaryotic gene which has been digested by the restriction endonuclease BamHI.



Following transformation, bacterial cells were grown in four different media, as shown below:

- I nutrient broth plus ampicillin
- II nutrient broth plus tetracycline
- III nutrient broth plus ampicillin and tetracycline
- IV nutrient broth without antibiotics

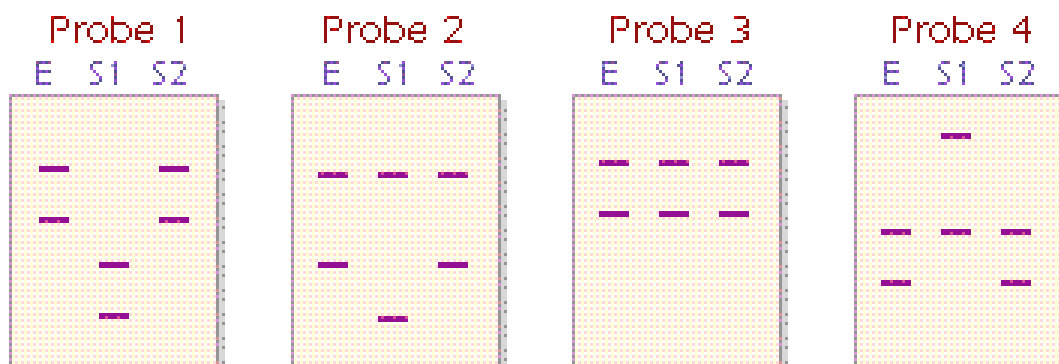
Which of the following media would bacterial cells that contain the recombinant plasmids grow in?

- A I and II
- B I and III
- C I and IV
- D IV only

Answer: C

NJC 07 Prelim

40 The figure below shows the results of an RFLP analysis of Evidence (E) taken from the murder scene matched against two suspects S1 and S2.



What is the conclusion of the investigation?

- A Probability that S1 was at the murder scene is low
- B Probability that S1 was at the murder scene is high
- C Probability that S2 was at the murder scene is low
- D Probability that S2 was at the murder scene is high

Answer: D